

PHARMACEUTICAL SECTOR

Portfolio Construction, Performance Analysis and Investor Recommendations

Research Report – Group Project (Portfolio Management)

Analysis Period: April 2021 – February 2026 (5 years)

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1. Project Objective and Sector Allocation

This project applies fundamental analysis, portfolio construction and performance-evaluation techniques to real market data from the Indian pharmaceutical sector. The group was allocated the Pharmaceuticals & Healthcare sector, selected for its structural growth drivers –India's position as a low-cost, high-volume generics and API manufacturer for global markets, a large and growing domestic branded-formulations market, and demand resilience through economic cycles. The sector also offers useful diversity for this analysis: export-oriented API/CRAMS players (e.g. Divi's Laboratories), domestic branded-formulation leaders (e.g. Alkem, Torrent), diversified generics majors (e.g. Sun Pharma, Dr Reddy's, Cipla, Lupin, Aurobindo, Zydus) and a high-return MNC subsidiary (Abbott India).

The objective is to construct and compare six alternative portfolios built from the same universe of 10 pharmaceutical companies (plus two comparable mutual funds and a blended benchmark), and to recommend suitable portfolios for conservative, moderate and aggressive investors based on quantitative performance measures and the underlying fundamentals.

2. Company and Mutual Fund Selection Rationale

Ten NSE-listed pharmaceutical companies were selected to represent the breadth of the sector by business model and market capitalisation: Sun Pharmaceutical Industries, Dr. Reddy's Laboratories, Cipla, Divi's Laboratories, Torrent Pharmaceuticals, Zydus Lifesciences, Lupin, Aurobindo Pharma, Alkem Laboratories and Abbott India. Together they span large-cap diversified generics, export-oriented API/CRAMS manufacturing, India-focused branded generics and an MNC subsidiary, giving a representative cross-section of the sector rather than a concentration in any single business model.

For the two comparable mutual fund schemes (Portfolios 5 and 6), the SBI Healthcare Opportunities Fund (Direct Plan – Growth) and Nippon India Pharma Fund (Direct Plan – Growth) were chosen. Both are large, well-established, pure-play sectoral funds with long track records (since 2013 and 2004 respectively) and substantial assets under management, making them suitable, credible comparators for the sector mandate.

3. Fundamental Analysis of the 10 Companies

Key profitability, growth and financial-strength indicators (FY2026, unless stated) are summarised below. All figures are drawn from the companies' audited financial statements (Profit & Loss, Balance Sheet and Cash Flow) for FY2022–FY2026.

Company	EBITDA Margin	ROE	Revenue Growth (FY26)	Revenue CAGR (FY22–26)	Net Profit CAGR (FY22–26)	Net Debt (₹ Cr)	EPS (FY26)
Sun Pharmaceutical Industries Ltd	33.3%	13.8%	11.2%	10.9%	35.8%	(6,976)	₹47.84
Dr. Reddy's Laboratories Ltd	25.3%	11.0%	3.2%	11.8%	17.5%	4,385	₹50.27
Cipla Ltd	25.2%	11.2%	2.2%	6.7%	11.0%	(693)	₹48.02
Divi's Laboratories Ltd	37.0%	15.3%	12.8%	4.2%	-3.5%	(3,407)	₹96.73
Torrent Pharmaceuticals Ltd	40.6%	25.5%	21.4%	13.2%	28.8%	(1,165)	₹63.92
Zydus Lifesciences Ltd	24.1%	18.9%	16.8%	15.5%	2.6%	11,079	₹50.09
Lupin Ltd	24.1%	23.9%	23.1%	14.3%	NM (loss in FY22)	963	₹116.65
Aurobindo Pharma Ltd	15.7%	9.2%	6.1%	9.4%	7.3%	(1,797)	₹60.34

Company	EBITDA Margin	ROE	Revenue Growth (FY26)	Revenue CAGR (FY22–26)	Net Profit CAGR (FY22–26)	Net Debt (₹ Cr)	EPS (FY26)
Alkem Laboratories Ltd	18.6%	17.0%	13.5%	8.4%	8.8%	314	₹192.51
Abbott India Ltd	23.5%	32.5%	8.1%	8.9%	18.1%	(2,083)	₹730.39

Net Debt shown in parentheses indicates a net cash position. Lupin's FY2022 net profit was negative (turnaround year), so a Net Profit CAGR is not meaningful (NM) and is excluded from ranking calculations.

Company Profiles and Investment Rationale

Sun Pharmaceutical Industries Ltd –India's largest pharmaceutical company by revenue, with a diversified generics and branded specialty portfolio (dermatology, ophthalmology, oncology) across the US, India and emerging markets.

Dr. Reddy's Laboratories Ltd –Global generics and API manufacturer with a strong US generics franchise, expanding into biosimilars and consumer wellness (Nicotex).

Cipla Ltd –Leader in respiratory/inhalation therapy and India branded generics, with a growing presence in the US and South Africa.

Divi's Laboratories Ltd –India's largest custom-synthesis (CRAMS) and API manufacturer, export-oriented with high margins and a blue-chip global innovator client base.

Torrent Pharmaceuticals Ltd –Strong branded-formulations franchise in India and Brazil with a cardiac/CNS focus; the sector's highest EBITDA margin and ROE among peers here.

Zydus Lifesciences Ltd –Diversified generics, specialty and consumer-wellness player with a growing US complex-generics and vaccines pipeline.

Lupin Ltd –Respiratory, anti-TB and cardiovascular generics major with a strong US and India presence; turned around from a FY2022 net loss to strong FY2026 profitability.

Aurobindo Pharma Ltd –Large-scale, vertically integrated generics manufacturer (API + formulations) with a growing biosimilars and vaccines pipeline.

Alkem Laboratories Ltd –India-focused branded-generics leader in acute therapies, with an expanding US generics business.

Abbott India Ltd –MNC subsidiary focused on branded formulations (women's health, gastro-intestinal, other segments); the sector's highest ROE and a near-debt-free balance sheet.

Key Observations

- Torrent Pharma and Abbott India post the sector's strongest profitability (EBITDA margin 40.6% and ROE 32.5% respectively), reflecting premium branded-formulation franchises with limited capital intensity.
- Divi's Laboratories and Torrent Pharma carry the strongest balance sheets among growth names (net cash positions), while Zydus Lifesciences and Dr. Reddy's carry net debt, linked to acquisition-funded expansion.
- Lupin, Torrent Pharma and Sun Pharma show the sharpest profit recovery/growth (net profit CAGR 28–36%) over FY2022–26, aided by a low FY2022 base and US generics price stabilisation.
- Valuation indicators (P/E, P/B) were assessed using each stock's market price as of 27 February 2026 against FY2026 EPS; premium names (Abbott India, Divi's Laboratories) trade at materially higher multiples than diversified generics majors, consistent with their higher ROE and earnings quality.

4. Data Sources, Assumptions and Methodology

Data Sources

- Company financials: audited annual financial statements, FY2022–FY2026 (Profit & Loss, Balance Sheet, Cash Flow).
- Daily stock prices: NSE historical data archives, 1 April 2021 to 27 February 2026, adjusted for corporate actions –specifically Dr. Reddy's Laboratories' 1:5 stock split (October 2024) and Torrent Pharmaceuticals' 1:2 stock split (July 2022), both back-adjusted so returns are computed on a like-for-like basis.
- Mutual fund NAV history: AMFI / fund-house NAV records, Direct Plan – Growth option, same period.
- Risk-free rate: 6.82% p.a., a representative assumption for 91-day Treasury Bill yields over the analysis period (which ranged approximately 2.9%–6.9%).

Methodology

- Return frequency: daily returns were computed for all 10 stocks and both mutual funds and annualised using 252 trading days per year, rather than monthly returns, to maximise the number of observations (≈ 1,220 daily data points per series) for more statistically stable covariance and beta estimates. The same frequency and annualisation method were applied consistently across all six portfolios.
- Market benchmark for Beta, Treynor Ratio and Jensen's Alpha: a true sector index (Nifty Pharma) was not accessible during this project, so the SBI Healthcare Opportunities Fund's return series was used as a consistent market proxy for all six portfolios. Portfolio 4 itself was separately constructed as an equally blended (50:50) daily return series of both mutual funds, representing a composite, professionally-managed sector reference for direct return/risk comparison alongside the six portfolios.
- Rebalancing: none. All portfolios use a buy-and-hold approach with weights fixed at the start of the period, consistently applied across Portfolios 1–3.
- Short selling and borrowing were not permitted in any portfolio; all weights are non-negative and sum to 100%.
- As portfolio weights for Portfolios 2 and 3 were both estimated and evaluated using the same five-year historical dataset, results should be read as an in-sample historical analysis rather than an out-of-sample forecast of future performance.

5. Portfolio Construction and Weighting Decisions

Portfolio 1 – Equally Weighted

10% allocated to each of the 10 companies –a naive-diversification baseline against which the more deliberate weighting schemes can be measured.

Portfolio 2 – Fundamentals-Weighted (Group's Own Criteria)

Weights were derived from a composite score combining each company's rank on Return on Equity, FY2026 Revenue Growth, and (inversely) Net Debt –rewarding profitable, fast-growing, financially conservative businesses. Scores were normalised to sum to 100%.

Portfolio 3 – Efficient Markowitz Portfolio

Using the daily return, variance and covariance data for the 10 stocks, an optimiser (Excel Solver–equivalent quadratic optimisation) was used to maximise the Sharpe Ratio subject to: weights summing to 100%, no short-selling, and –to avoid an unrealistic corner solution concentrated in one or two stocks –a maximum weight of 25% per stock. The resulting efficient portfolio is used for comparison.

Company	Portfolio 1 Weight	Portfolio 2 Weight	Portfolio 3 Weight
Sun Pharmaceutical	10.00%	8.48%	25.00%
Dr. Reddy's Labs	10.00%	16.36%	3.00%
Cipla	10.00%	14.55%	3.29%
Divi's Labs	10.00%	7.88%	8.59%
Torrent Pharma	10.00%	5.45%	10.62%
Zydus Lifesciences	10.00%	10.30%	15.77%
Lupin	10.00%	7.27%	0.00%
Aurobindo Pharma	10.00%	13.33%	1.24%
Alkem Labs	10.00%	9.70%	19.12%
Abbott India	10.00%	6.67%	13.38%

Portfolio 4 – Benchmark Portfolio

Constructed as an equally blended (50:50) daily return series of the SBI Healthcare Opportunities Fund and Nippon India Pharma Fund, used as the sector benchmark in place of an inaccessible market index (see Section 4).

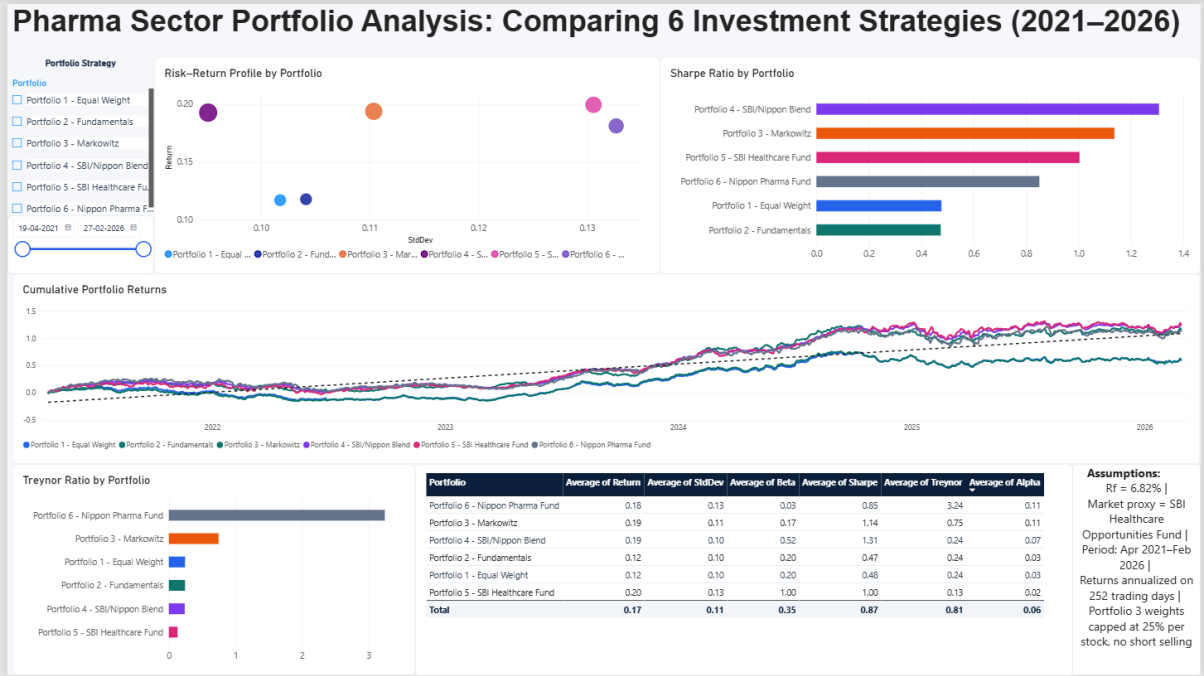
Portfolio 5 & 6 – Comparable Mutual Fund Schemes

SBI Healthcare Opportunities Fund (Direct – Growth) and Nippon India Pharma Fund (Direct – Growth) –both pure-play, large, long-track-record sectoral funds whose investment mandate closely matches the assigned pharmaceutical sector (see Section 2 for selection rationale).

6. Comparative Performance Analysis

All six portfolios were compared over the common five-year period (April 2021–February 2026) using annualised return, annualised standard deviation, Beta (against the SBI Healthcare Opportunities Fund), Sharpe Ratio, Treynor Ratio and Jensen's Alpha (risk-free rate: 6.82% p.a.).

Portfolio	Ann. Return	Ann. Std Dev	Beta	Sharpe	Treynor	Alpha
Portfolio 1 – Equally Weighted	11.68%	10.18%	0.20	0.48	0.24	2.63%
Portfolio 2 – Fundamentals-Weighted	11.76%	10.41%	0.20	0.47	0.24	2.66%
Portfolio 3 – Markowitz (Max Sharpe)	19.37%	11.04%	0.17	1.14	0.75	10.67%
Portfolio 4 – Blended Benchmark (SBI+Nippon)	19.26%	9.51%	0.52	1.31	0.24	6.61%
Portfolio 5 – SBI Healthcare Opportunities Fund	19.93%	13.06%	1.00	1.00	0.13	1.94%
Portfolio 6 – Nippon India Pharma Fund	18.11%	13.27%	0.03	0.85	3.24	10.90%



Dashboard: risk-return scatter, Sharpe/Treynor comparison and cumulative growth of the six portfolios (Power BI).

Why Portfolio Rankings Differ Across Measures

The six portfolios rank differently depending on which measure is used, and this is expected rather than an inconsistency:

- By raw annualised return, the SBI Healthcare Fund (19.93%) narrowly leads Portfolio 3 (19.37%) and the blended benchmark (19.26%) –all three far ahead of the naive Portfolios 1 and 2 (≈11.7%).
- By total risk (standard deviation), the blended benchmark (Portfolio 4) is the least volatile (9.51%) because averaging two only moderately correlated funds diversifies away idiosyncratic fund-manager risk; the individual mutual funds (Portfolios 5 and 6) are the most volatile (≈13%) since they carry full single-scheme concentration risk.
- The Sharpe Ratio, which uses total risk, therefore favours Portfolio 4 (1.31) and Portfolio 3 (1.14) – both combine strong returns with comparatively low volatility.
- The Treynor Ratio and Jensen's Alpha use Beta (systematic risk relative to the SBI benchmark) rather than total risk. Portfolio 6 (Nippon Pharma Fund) shows an unusually high Treynor Ratio (3.24) and Alpha (10.90%) almost entirely because its Beta against SBI is very low (0.03) –not because of exceptional risk-adjusted skill. Dividing a healthy excess return by a near-zero Beta mechanically inflates the Treynor Ratio; this should be treated as a limitation of using a single mutual fund as the market proxy (see Section 8) rather than read at face value.
- Excluding that outlier, Portfolio 3 (Markowitz) has the next-strongest Treynor Ratio (0.75) and a high Alpha (10.67%), reinforcing that mean-variance optimisation meaningfully improved risk-adjusted performance over the naive Portfolios 1 and 2, whose Alphas are modest (≈2.6%).

7. Recommendations for Investor Profiles

Conservative Investor

Portfolio 4 (Blended Benchmark: SBI Healthcare + Nippon Pharma, 50:50) is the most suitable among the six. It combines the lowest total risk (9.51% standard deviation) with a strong return (19.26%) and the highest Sharpe Ratio (1.31), because blending two funds diversifies away single-manager and single-scheme concentration risk. That said, no portfolio here is conservative in an absolute sense –all six carry double-digit or near-double-digit annualised volatility and full sector concentration in pharmaceuticals. A genuinely

conservative investor should treat any of these as a satellite/sector allocation within a broader, more diversified portfolio rather than a core low-risk holding.

Moderate Investor

Portfolio 3 (Markowitz-Optimal) is recommended. It delivers a return (19.37%) comparable to the best-performing options with only marginally higher risk than Portfolio 1/2 (11.04% vs. ≈10%), the lowest Beta among growth-oriented options (0.17), and the second-highest Sharpe Ratio (1.14) and Alpha (10.67%) after excluding the Portfolio 6 outlier –a well-balanced trade-off between growth and risk grounded in optimisation rather than a fixed rule.

Aggressive Investor

The SBI Healthcare Opportunities Fund (Portfolio 5) is recommended for investors prioritising maximum long-term growth and willing to accept the highest volatility (13.06% –the highest of all six portfolios) for the highest absolute return (19.93%). The Nippon India Pharma Fund (Portfolio 6) is a reasonable alternative given its high Alpha, but its performance-measure outlier (Section 6) and reliance on active management concentrated in a single scheme warrant caution before treating its risk-adjusted figures at face value.

8. Conclusion, Limitations and References

Conclusion

Over the five-year period, the Markowitz-optimised Portfolio 3 clearly outperformed the naive Portfolios 1 and 2 on a risk-adjusted basis, validating the value of systematic optimisation over simple heuristics. The professionally managed mutual funds (Portfolios 5 and 6) delivered the highest absolute returns but with correspondingly higher volatility, while blending them (Portfolio 4) produced the best overall risk-return trade-off. Fundamentally, the sector's strongest businesses –Torrent Pharma, Abbott India and Divi's Laboratories – combine high profitability with resilient balance sheets, supporting the sector's inclusion as a satellite growth allocation for investors across the risk spectrum.

Limitations

- A true sector index (Nifty Pharma) could not be sourced; a mutual fund (SBI Healthcare Opportunities Fund) was used as the market proxy for Beta, Treynor Ratio and Jensen's Alpha. Since fund returns already embed manager skill, this understates true systematic (market) risk and can distort Treynor/Alpha for portfolios weakly correlated with that specific fund (notably Portfolio 6).
- Daily returns (annualised) were used instead of the monthly returns recommended in the project brief, chosen to increase the number of observations for covariance estimation; results may differ modestly under a monthly-return methodology.
- The analysis is in-sample: Portfolio 2 and Portfolio 3 weights were both derived from and evaluated against the same five-year dataset, so reported performance reflects historical fit rather than a genuine out-of-sample forecast.
- No portfolio rebalancing, transaction costs, taxes, or mutual fund expense ratios were incorporated into the return calculations.

References

- National Stock Exchange of India (NSE) – historical daily price data, 10 constituent companies, 1 April 2021 to 27 February 2026.
- Company audited annual financial statements, FY2022–FY2026 (Profit & Loss, Balance Sheet, Cash Flow).
- Association of Mutual Funds in India (AMFI) / respective fund houses – NAV history, SBI Healthcare Opportunities Fund and Nippon India Pharma Fund (Direct Plan – Growth), same period.
- Reserve Bank of India (RBI) – 91-day Treasury Bill yield data, used for the risk-free rate assumption.